

Prevalence and public health significance of aluminum residues in milk and some dairy products.



Sixty random samples of bulk farm milk, market milk, locally manufactured processed cheese, and milk powder were collected to be analyzed for aluminum (Al) concentration using graphite furnace atomic absorption spectrometry (GFAAS). The results were compared with provisional acceptable permissible limits (PAPLs). The maximum estimated dietary intake (MEDI) of Al for the examined samples was calculated. In addition, an experimental study was conducted to determine the possible leaching of Al from cookware in milk during boiling. The obtained results showed that Al concentration in examined bulk farm milk samples was found to be negligible. In contrast, market milk revealed higher concentration, 65.0% of the examined samples were above the PAPLs. The results revealed significant difference of Al concentration among them. The Al levels in processed cheese wrapped in Al foil were significantly higher than those found in samples packed in glass containers with a significant difference of Al concentration between them. Also, 20% of the examined milk powder samples exceeded the PAPLs (0.01 to 0.4 mg/kg). The MEDI for Al in bulk farm milk, control market milk, market milk boiled in Al cookware, market milk boiled in stainless-steel cookware, processed cheese wrapped in Al foil, processed cheese packed in glass containers, and milk powder were calculated as 3.0%, 61.0%, 63.0%, 61.0%, 428.0%, 220.0%, and 166.0% from "PTDI," respectively. The results of the experimental study showed no marked significant differences of Al concentration between market milk (control group) and those boiled in Al cookware, as well as to those boiled in stainless-steel cookware. PRACTICAL APPLICATION: The results of the present study indicate that Al level in milk kept in Al containers and dairy products packed in Al foil is beyond the permissible limits, suggesting health hazard. Therefore, all milk cans should be constructed of stainless steel, prevent the entrance of tap water into milk, and the processed cheese should be packed in glass containers and not wrapped in Al foil. Leaching of Al increased to a significant percent more during storage than during boiling, so milk should be kept in stainless steel or glass containers in the refrigerator.